



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



UNC 1549 Threat Actor Profile

Threat Actor Profile

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Published	2026-06-24
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Author	Lost Boys Cyber
Report Type	Threat Actor Profile

TLP:CLEAR | Lost Boys Cyber | UNC 1549 Threat Actor Profile - Threat Actor Profile

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1. Executive Summary

Field	Assessment
Report title	UNC1549 Threat Actor Profile
Report family	Threat Actor Profile
Classification	TLP:CLEAR
Prepared for	Lost Boys Cyber review workflow
Date	2026-06-24
Primary actor names	UNC1549; Screening Serpens; Smoke Sandstorm / BOHRIUM overlap; Iranian Dream Job / TA455 reporting overlap
Primary motivation	Strategic espionage aligned with Iranian state intelligence requirements
Primary sectors	Aerospace, aviation, defense, defense manufacturing, telecommunications, technology, IT and engineering personnel
Primary regions	Middle East, especially Israel and UAE; reported expansion into Western Europe, the United States, and additional Middle Eastern targets
Confidence	Moderate to high for activity cluster and TTPs; moderate for organizational attribution to Iranian state organs / IRGC-linked ecosystems

UNC1549 is an Iran-nexus cyber-espionage cluster best understood as a patient access-development and intelligence-collection actor targeting the defense industrial base and adjacent technology supply chains. Public reporting from Mandiant, Unit 42, ClearSky, Check Point, and secondary security press depicts a group active since at least 2022 that pairs highly tailored recruitment-themed social engineering with cloud-hosted command-and-control, DLL search-order hijacking, signed or sideloaded malware, credential theft, and long-lived persistence. Mandiant assesses UNC1549 activity with moderate confidence as tied to an Iran-based espionage group overlapping Tortoiseshell and Smoke Sandstorm / BOHRIUM; Unit 42 tracks the broader actor as Screening Serpens and assesses recent 2026 activity with moderate-high confidence.

The actor's operational culture is pragmatic and adaptive rather than noisy. UNC1549 is willing to spend time learning a target's business relationships, job-seeking patterns, internal password reset workflows, source code, help desk and ticketing processes, and third-party access pathways. That human and organizational reconnaissance is then converted into believable lures, lookalike infrastructure, and post-compromise credential operations. The group has repeatedly used Microsoft Azure-hosted infrastructure and victim-industry naming patterns to blend command-and-control with normal cloud traffic. In 2025 reporting, Mandiant described UNC1549 planting backdoors that remained quiet for months and only activated after victim remediation, a strong signal that the group anticipates incident response and designs for re-entry.

UNC1549 matters because its objectives map directly to strategic intelligence and operational leverage. Aerospace and defense networks contain export-controlled engineering data, sustainment documentation, program plans, supplier diagrams, privileged remote-access paths, and email context that can support espionage, procurement, influence, counter-sanctions activity, and potentially kinetic targeting. Telecommunications and technology providers offer a second-order benefit: access into customers, partners, and operational communications. Defenders should treat UNC1549 as a supply-chain and identity adversary as much as a malware adversary. Blocking a single hash or Azure domain is useful but insufficient; the durable defense is strong third-party access governance, identity telemetry, DLL sideloading detection, AppDomainManager hijack detection, outbound tunnel monitoring, and rapid credential invalidation after suspected exposure.

Key analytic judgments:

- UNC1549 is not a single-tool malware problem. It is a blended identity, social-engineering, third-party-access, and custom-tooling threat.
- The actor has evolved from MINIBIKE / MINIBUS / LIGHTRAIL tradecraft into a broader tool ecosystem that includes TWOSTROKE, DEEPROOT, GHOSTLINE, POLLBLEND, DCSYNCR.SLICK, CRASHPAD, SIGHTGRAB, TRUSTTRAP, MiniUpdate, and MiniJunk V2 reporting.
- Azure Web Apps, cloudapp.azure.com hosts, ONLYOFFICE / file-sharing delivery, spoofed recruiting portals, and meeting-platform lures should be hunted as behavioral patterns, not only static IOCs.
- The most defensible detection opportunities are DLL search-order hijacking, unusual .NET application configuration files that disable ETW or abuse AppDomainManager, scheduled-task persistence under user profiles, reverse SSH tunnels on TCP/443, DCSync behavior from non-standard hosts or newly manipulated computer accounts, and suspicious third-party VDI / Citrix / VMware / AVD session behavior.

2. Intelligence Requirement

This profile answers the following intelligence requirements for Lost Boys Cyber defensive planning:

Requirement	Analytic answer
Who is UNC1549?	A suspected Iran-nexus espionage actor / cluster overlapping public Smoke Sandstorm, BOHRIUM, Tortoiseshell, Screening Serpens, and Iranian Dream Job reporting.
What does UNC1549 want?	Sensitive technical, defense, aviation, telecommunications, and identity data; durable access into primary targets and their partners; credentials enabling lateral movement and future operations.
Who is at risk?	Aerospace, aviation, defense manufacturing, telecommunications, technology, managed service providers, suppliers, and personnel with engineering / IT / privileged access.
How does UNC1549 gain access?	Tailored spear phishing, fake job portals, credential-harvesting pages, meeting-platform impersonation, compromised third-party accounts, trusted relationship abuse, and VDI / Citrix / VMware / AVD breakout.
What tradecraft should defenders prioritize?	DLL search-order hijacking, cloud-hosted C2, reverse SSH tunnels, .NET AppDomainManager hijacking, scheduled-task persistence, browser credential theft, DCSync / RBCD / AD CS abuse, and fake credential prompts.
What should be hunted now?	Recruitment-themed payload chains; Azure Web App C2 naming patterns; unusual scheduled tasks; OpenSSH reverse tunnels; DCSync from non-DC or suspicious DC-account contexts; new AppDomainManager XML directives; sideloading near VMware, Citrix, Fortinet, Microsoft, NVIDIA, and conferencing binaries.

Assumptions and scope:

- This report uses public intelligence available as of 2026-06-24 and does not rely on private telemetry from a victim environment.
- The report treats UNC1549 / Screening Serpens as the central analytic object while noting that vendor naming may not map one-to-one across all campaigns.
- Actor attribution to Iranian state organs is assessed from public reporting and alignment with strategic interests; the report does not claim legal attribution beyond cited vendor assessments.
- IOCs are included for enrichment and triage, but recommended defenses emphasize behaviors because UNC1549 rotates infrastructure and customizes payloads per victim.

3. Source Review and Confidence

Source	Date	Value	Confidence contribution	Limitations
Google Cloud / Mandiant, “When Cats Fly”	2024-02-27	Foundational UNC1549 reporting: attribution, targeting, MINIBIKE, MINIBUS, LIGHTRAIL, Azure C2, fake job and Israel-Hamas lures, IOCs.	High for 2022-2024 tradecraft and IOCs.	Vendor visibility; moderate-confidence attribution; some infrastructure now stale.
Google Cloud / Mandiant, “Frontline Intelligence”	2025-11-17	Incident-response-derived TTPs: third-party access, VDI breakouts, TWOSTROKE, DEEPROOT, DCSYNCR.SLICK, CRASHPAD, SIGHTGRAB, TRUSTTRAP, reverse SSH, MITRE mapping.	High for observed TTPs; high operational relevance.	Some IOCs are gated in GTI or may be selectively published.
Unit 42, “Tracking Iranian APT Screening Serpens’ 2026 Espionage Campaigns”	2026-05-22	Recent evolution: MiniUpdate, MiniJunk V2, AppDomainManager hijacking, personalized recruiting and meeting lures, 2026 targets, IOCs.	High for malware-analysis detail; moderate-high for cluster attribution per source.	Vendor naming uses Screening Serpens; mapping to UNC1549 depends on overlap assessment.
ClearSky, “Iranian Dream Job” campaign	2024-11	Independent reporting on Iranian Dream Job / TA455-style recruitment lures and SnailResin / SlugResin ecosystem.	Medium; useful for lure and alias context.	Public PDF details may overlap adjacent Iranian clusters, not necessarily every UNC1549 intrusion.
Check Point reporting on Nimbus Manticore / Western Europe expansion	2025	Supports actor expansion into Western Europe and details code-signing / sideloaded patterns referenced by other vendors.	Medium to high for corroboration.	Naming and clustering may differ from Mandiant / Unit 42.
Secondary security press	2024-2026	Useful for tracking public summaries and wider industry interpretation.	Low to medium; corroborative only.	Often rephrases primary vendor research.

Analytic confidence:

Judgment	Confidence	Rationale
UNC1549 conducts espionage against aerospace, aviation, defense,	High	Multiple independent reports and incident-response observations align on sector

telecommunications, and technology-related targets.		targeting.
UNC1549 is Iran-nexus and overlaps Smoke Sandstorm / BOHRIUM / Tortoiseshell / Screening Serpens reporting.	Moderate to high	Vendor assessments converge, but naming boundaries differ and public evidence is not complete.
The group is increasingly identity- and supply-chain-centric.	High	2025 Mandiant reporting describes trusted relationships, third-party VDI, service-provider pivots, ticketing abuse, and credential operations.
Azure-hosted and victim-themed infrastructure is a persistent operational pattern.	High	Repeated across MINIBIKE, MINIBUS, LIGHTRAIL, TWOSTROKE, DEEPROOT, POLLBLEND, MiniUpdate, and MiniJunk V2 reporting.
The actor will continue adapting initial access lures to current regional events and professional workflows.	High	Historical use of job lures, Israel-Hamas themes, meeting impersonation, and recruiter-style targeting shows rapid theme adaptation.

4. Key Findings

ID	Finding	Defensive implication	Confidence
KF-01	UNC1549 blends social engineering with trusted relationship abuse. Fake job portals remain important, but compromised third-party credentials and VDI / Citrix / VMware / Azure Virtual Desktop access are now central.	Audit all partner access, enforce phishing-resistant MFA, isolate VDI sessions, monitor breakouts, and treat vendor accounts as high-risk identities.	High
KF-02	The actor's malware is custom but not static. MINIBIKE and MINIBUS evolved into a larger backdoor and utility ecosystem, while 2026 reporting adds MiniUpdate and MiniJunk V2.	Do not anchor detection solely to old hashes; build detections around sideloading, scheduled tasks, C2 patterns, and loader behavior.	High
KF-03	UNC1549 repeatedly uses Azure Web Apps / cloudapp.azure.com and industry-themed domains to make malicious traffic look business-relevant.	Proxy, DNS, and EDR detections should score new Azure-hosted domains by naming pattern, process lineage, rarity, and tenant/user context.	High
KF-04	Post-compromise activity emphasizes credentials: browser credential theft, fake login prompts, DCSync, Kerberoasting, RBCD, AD CS abuse, password resets for domain-controller computer accounts, and privileged	Identity teams need AD replication-right monitoring, certificate-template review, domain-controller account change alerts, and rapid password/session invalidation playbooks.	High

	password-reset phishing.		
KF-05	UNC1549 anticipates defenders. Mandiant observed quiet backdoors, delayed activation after remediation, artifact deletion, RDP history clearing, and reverse SSH tunnels that reduce host artifacts.	Eradication plans should include full identity reset, partner credential reset, cloud/API token review, reverse-tunnel sweeps, and follow-up monitoring for dormant access.	High
KF-06	Newer 2026 tradecraft includes .NET AppDomainManager hijacking and native configuration directives that disable or reduce telemetry, including ETW-related controls.	Detection engineering should parse application .config files, flag AppDomainManager usage outside known-good apps, and detect ETW disablement directives.	Medium to high
KF-07	UNC1549's targeting choices align with Iranian strategic interests around aerospace, defense, telecommunications, and regional conflicts.	Executives should treat this as strategic intelligence collection, not opportunistic commodity intrusion.	Moderate to high

4.1. Actor narrative and mindset

UNC1549 behaves like an operator set tasked with getting close to people and systems that hold national-security-relevant knowledge. The group is comfortable with deception at multiple layers: fake recruiters, fake job portals, cloned login pages, spoofed meeting invitations, decoy applications, code-signed or sideloaded payloads, cloud infrastructure with innocent naming, and malicious activity launched from legitimate third-party sessions. The actor's mindset is opportunistic inside a strategic frame: if a direct defense contractor is hard to breach, compromise a supplier; if malware is detected, leave dormant access; if endpoint controls are strong, abuse signed binaries, DLL search-order hijacking, AppDomainManager configuration, and reverse SSH.

The group's tradecraft shows a practical understanding of how defense and technology organizations actually work. Recruitment processes are noisy, engineers receive external files, suppliers need remote access, VDI boundaries are often trusted too much, password reset portals are familiar, and Azure traffic is rarely blocked outright. UNC1549 exploits those realities rather than relying only on zero-day exploitation. This makes user education necessary but insufficient: the actor's best opportunities are often in identity governance, supplier trust, remote access design, and incomplete remediation.

5. Threat Context

5.1. Strategic background

UNC1549's publicly reported targeting aligns with Iranian intelligence priorities: defense procurement, aerospace and aviation technology, telecommunications access, regional adversary visibility, and strategic situational awareness during periods of conflict. Mandiant reported targeting of Israel and the UAE and potential targeting or interest involving Turkey, India, and Albania. Later reporting expands the target set to Western Europe, the United States, and additional Middle Eastern entities. The actor's interest in aerospace, aviation, and defense is particularly consequential because those networks often include military-adjacent engineering, supplier diagrams, export-controlled intellectual property, facility access processes, logistics data, and privileged remote connectivity to partners.

5.2. Victimology and targeting

Target category	Why it matters to UNC1549	Common access angle
Aerospace and aviation firms	Engineering, flight systems, supplier data,	Fake job offers, meeting lures, supplier

	operational plans, export-controlled technology.	access, Azure-themed C2.
Defense manufacturers and contractors	Military programs, procurement, sensitive documents, partner ecosystems.	Third-party partner accounts, VDI / Citrix / VMware / AVD, credential theft.
Telecommunications providers	Critical infrastructure, customer data, communications metadata, strategic access into other targets.	LinkedIn / recruitment lures, MINIBIKE-style malware, credential harvesting.
Technology and IT personnel	Privileged access, source code, identity systems, admin tooling, network diagrams.	Role-relevant phishing, password reset impersonation, fake meeting portals.
Suppliers and MSP-like partners	Lower-friction route into mature primary targets.	Trusted relationship abuse, compromised vendor credentials, remote access portals.
Individuals seeking employment	High receptivity to job documents, portals, and recruiter contact.	Personalized recruitment lures, ONLYOFFICE / file-sharing payload delivery.

5.3. Campaign and tooling timeline

Period	Activity / tooling	Notes
2022	MINIBIKE early versions; LIGHTRAIL early tunneler; fake recruiting ecosystem.	MINIBIKE initially used C++ backdoor functionality, IMG/ZIP delivery, search-order hijacking, Run key persistence, and later Azure C2.
2023	MINIBIKE v2, MINIBUS, LIGHTRAIL v2, credential-harvesting pages, Israel-Hamas themed lures.	Mandiant observed fake job pages and “Bring Them Home Now” themed decoys.
2024	Continued public reporting on UNC1549 and Iranian Dream Job activity.	Industry awareness increased; actor retained recruitment-themed operations.
Late 2023-2025	Third-party access, VDI breakout, reverse SSH tunnels, DCSYNCR.SLICK, TWOSTROKE, DEEPROOT, CRASHPAD, SIGHTGRAB, TRUSTTRAP, GHOSTLINE, POLLBLEND.	Mandiant IR reporting shows a mature post-compromise playbook and long-term persistence.
Late 2025	Reported strategic expansion into Western Europe.	Check Point / Unit 42 context highlights broader operational reach.
Feb-Apr 2026	MiniUpdate and MiniJunk V2; AppDomainManager hijacking; job and meeting lures; target indications in U.S., Israel, UAE, and other Middle Eastern entities.	Unit 42 reported six new RAT variants and high operational tempo.

5.4. Initial access and delivery chains

Chain	Observed components	Detection focus
Fake job / recruiting portal	LinkedIn or email contact; fake recruiter or job page; PDFs / ZIP archives; nested	External-link clicks, archive downloads, unsigned DLLs near signed setup binaries,

	“Hiring Portal.zip”; malicious setup.exe; decoy errors or portals.	hidden files in archives, newly created scheduled tasks.
Credential harvesting	Fake Boeing / Teledyne FLIR-like pages, password reset pages, vendor or partner login themes, privileged-account phishing after mailbox reconnaissance.	Brand-lookalike domains, impossible-travel or new-device logins, OAuth / session anomalies, password reset telemetry.
Third-party trusted relationship	Compromised partner accounts; Citrix / VMware / AVD / VDI access; breakout to underlying host or adjacent networks.	Vendor account risk scoring, VDI clipboard/file-transfer anomalies, child process creation from virtualized sessions, unusual network pivots.
Meeting-platform impersonation	Authentic meeting links used to build trust; lookalike meeting page; hidden JavaScript download redirect; fake installer.	Newly registered meeting domains, downloads from file-sharing services, process lineage from meeting installers.
ONLYOFFICE / file-sharing payload delivery	Attacker-managed document storage hosting payload archives.	File-sharing URL reputation, archives with hidden DLLs, endpoint execution from Downloads/Temp.

5.5. Malware and tool profile

Tool / family	Platform	Role	Notable behaviors
MINIBIKE	Windows / C++	Backdoor	File enumeration, upload/download, command execution, Azure C2, search-order hijacking, OneDrive / SharePoint / decoy app lures, Run key persistence.
MINIBUS	Windows	Backdoor	More flexible command interface, process enumeration, Azure plus dedicated domains, “Bring Them Home Now” decoy, OneDriveCoUpdate persistence.
LIGHTRAIL	Windows	Tunneler	Socks4a / Lastenzug-like tunneler, WebSocket / Azure communication, fixed user agent, DLL export lastenzug.dll.
TWOSTROKE	Windows / C++	Backdoor	SSL-encrypted TCP/443, bot ID generation, file operations, DLL loading, command execution.
DEEPROOT	Linux / Go	Backdoor	Shell execution, system info, file listing/delete/upload/download, multiple Azure-hosted C2s.
GHOSTLINE	Windows / Go	Tunneler	go-yamux-based network tunneling.

POLLBLEND	Windows / C++	Tunneler	Registration and configuration endpoints, HTTP POST registration pattern.
DCSYNCER.SLICK	Windows	Credential theft	DCSync-derived utility based on open-source / Mimikatz concepts, dynamic API resolution, LOG.txt output paths, used with manipulated DC computer accounts.
CRASHPAD	Windows / C++	Credential utility	DPAPI / CryptUnprotectData-related credential extraction, config.txt to crash.log patterns.
SIGHTGRAB	Windows / C	Collection	Periodic screenshots saved under Public Videos or Music timestamped directories.
TRUSTTRAP	Windows	Credential harvesting	Fake Outlook-style prompt captures credentials in cleartext.
MiniUpdate	Windows / .NET	RAT	AppDomainManager hijacking, ETW-related evasion, scheduled task persistence, UpdateChecker.dll, 16-18 opcode dispatcher, chunked exfiltration.
MiniJunk V2	Windows / .NET	RAT / loader ecosystem	DLL sideloading, AppDomainManager usage, junk-string padding, ONLYOFFICE delivery, C2 endpoints /api/app/check, /api/app/update, /api/app/comment.

5.6. Infrastructure patterns and observables

UNC1549 infrastructure is notable for blending into legitimate business and cloud naming conventions. Observed patterns include:

- Azure Web Apps domains with recruiting, survey, engineering, airline, API, health, finance, Windows-service, or IT-administration themes.
- cloudapp.azure.com hosts under regions such as eastus, westus3, northeurope, qatarcentral, centralus, and uaenorth.
- Dedicated domains that mimic business services, travel, finance, storage, developer tooling, or political / industry themes.
- File-sharing and collaboration platforms used as delivery staging, including ONLYOFFICE DocSpace and filemail-style URLs in 2026 reporting.
- Reverse SSH tunnel infrastructure on TCP/443 with OpenSSH options suppressing known-hosts prompts and enabling remote port forwarding.

Selected public IOCs for enrichment:

Type	Indicator	Context
Domain	1stemployer[.]com	Fake recruiting site associated with MINIBUS / UNC1549 overlap.
Domain	birngthemhomenow[.]co[.]il	Israel-Hamas themed lure / MINIBUS

		context.
Domain	cashcloudservices[.]com	MINIBUS-related infrastructure.
Domain	xboxplayservice[.]com	MINIBUS-related infrastructure.
Domain	airbus.usa-careers[.]com	2025 phishing domain for initial access.
Domain	airplaneserviceticketings[.]com	2025 MINIBIKE context.
Domain	politicalanorak[.]com	SSH tunneling context.
Domain	licencemanagers[.]azurewebsites[.]net	MiniJunk V2 / 2026 Unit 42 reporting.
Domain	NanoMatrix[.]azurewebsites[.]net	MiniJunk V2 / 2026 Unit 42 reporting.
Domain	PremierHealthAdvisory[.]com	MiniUpdate / 2026 Unit 42 reporting.
Domain	Ramiltonsfinance[.]com	MiniUpdate / 2026 Unit 42 reporting.
IP	104.194.215[.]88	SSH tunneling context in Mandiant 2025 reporting.
IP	13.60.50[.]172	SSH tunneling context in Mandiant 2025 reporting.
IP	167.172.137[.]208	SSH tunneling context in Mandiant 2025 reporting.
IP	34.18.42[.]26	SSH tunneling context in Mandiant 2025 reporting.
Hash MD5	691d0143c0642ff783909f983ccb8ffd	MINIBIKE v2 ZIP archive example.
Hash MD5	ef262f571cd429d88f629789616365e4	MINIBUS bringthemhomenow.zip example.
Hash SHA256	0db36a04d304ad96f9e6f97b531934594cd95a5cea9ff2c9af249201089dc864	MiniUpdate UpdateChecker.dll, U.S. campaign.
Hash SHA256	43dc62cef52ebdd69e79f10015b3e13890f26c058325c0ff139c70f8d8eadcfa	MiniJunk V2 Connection.dll, U.S. campaign.

5.7. MITRE ATT&CK mapping

Tactic	Technique	ID	UNC1549 behavior
Reconnaissance	Phishing for Information	T1598 / T1598.003	Credential-harvesting pages and password-reset themed lures.
Initial Access	Phishing	T1566	Job lures, meeting-platform impersonation, malicious archives and links.
Initial Access	Valid Accounts	T1078	Compromised partner / third-party credentials used for access.
Initial Access	Trusted Relationship	T1199	Pivoting from suppliers and

			partners into high-maturity targets.
Execution	User Execution	T1204	Victims execute job portal, meeting, or decoy installer payloads.
Persistence	Registry Run Keys / Startup Folder	T1060 / T1547.001	MINIBIKE and MINIBUS Run key persistence.
Persistence	Scheduled Task/Job	T1053.005	MiniUpdate / MiniJunk scheduled-task persistence.
Privilege Escalation	Abuse Elevation Control Mechanism	T1548	MiniUpdate UAC elevation capability; AD CS abuse for higher privilege.
Defense Evasion	DLL Search Order Hijacking	T1574.001	CRASHPAD, DCSYNCR.SLICK, GHOSTLINE, LIGHTRAIL, MINIBIKE, POLLBLEND, SIGHTGRAB, TWOSTROKE.
Defense Evasion	Signed Binary Proxy Execution / trusted binary abuse	T1218 family	Legitimate setup / VMware / Citrix / Microsoft / NVIDIA / conferencing binaries used to sideload payloads.
Defense Evasion	Indicator Removal	T1070	Utility deletion and RDP history registry cleanup.
Credential Access	OS Credential Dumping: DCSync	T1003.006	DCSYNCR.SLICK and DC computer-account manipulation.
Credential Access	Kerberoasting	T1558.003	Obfuscated Invoke-Kerberoast reported by Mandiant.
Credential Access	Credentials from Web Browsers	T1555.003	CRASHPAD / browser credential theft and RDP session hijacking.
Discovery	Account Discovery / Domain Groups	T1087 / T1069	net user, net group, AD Explorer, internal reconnaissance.
Lateral Movement	Remote Services: RDP	T1021.001	RDP used frequently for movement.
Lateral Movement	Windows Remote Management	T1021.006	PowerShell Remoting observed.
Lateral Movement	Remote Services / Remote Access Software	T1219	AWRC, SCCM remote control / SCCMVNC, ZeroTier, ngrok.
Command and Control	Web Service	T1102	Azure Web Apps and cloud-hosted C2.
Command and Control	Proxy / External Proxy	T1090	LIGHTRAIL, GHOSTLINE,

			POLLBLEND, reverse SSH tunnels.
Exfiltration	Exfiltration Over C2 Channel	T1041	Backdoor upload and chunked exfiltration features.
Collection	Screen Capture	T1113	SIGHTGRAB screenshot collection.
Collection	Data from Information Repositories	T1213 / T1213.002	Browsing and collecting Teams / SharePoint content.

6. Detection and Hunting

6.1. Detection priorities

Priority	Analytic	Why it matters	Data sources
1	DLL sideloading from legitimate vendor directories	Central execution method across UNC1549 tooling.	EDR process/module telemetry, Sysmon Event ID 7, file creation, code-signing data.
2	.NET AppDomainManager hijacking and telemetry-disabling .config files	Important 2026 evolution; can blind .NET telemetry before payload starts.	File telemetry, EDR, Sysmon Event ID 11/15, script/file-content scanning.
3	Reverse SSH tunnels on TCP/443	Mandiant observed extensive reverse SSH use limiting host artifacts.	Process command line, network telemetry, proxy, EDR.
4	Suspicious scheduled tasks under user context	MiniUpdate / MiniJunk persistence and immediate execution.	Windows Task Scheduler Operational logs, Security 4698/4702, EDR.
5	Azure Web Apps / cloudapp.azure.com C2 from unusual processes	Persistent infrastructure pattern.	DNS, proxy, netflow, EDR process-to-network mapping.
6	DCSync / replication-right anomalies	High-impact credential access.	DC Security logs 4662/4742/4724, AD replication telemetry, EDR.
7	Third-party VDI / Citrix / VMware / AVD breakout	Initial access and lateral movement vector against mature targets.	VDI logs, identity provider logs, EDR child process trees, network segmentation logs.
8	Browser credential theft and fake credential prompts	Supports privilege escalation and lateral movement.	File access to browser credential stores, DPAPI calls, suspicious GUI prompts, command line.

6.2. Sigma detection: AppDomainManager and ETW-disabling .NET configuration

```

title: Suspicious .NET AppDomainManager Hijack Configuration
id: 6bbcfefe-unc1549-appdomainmanager-config
status: experimental
description: Detects creation or modification of .NET application configuration files containing AppDomainManager hijack or telemetry-disabling directives associated with Screening Serpens / UNC1549 2026 tradecraft.

```

```

author: Lost Boys Cyber
references:
  - https://unit42.paloaltonetworks.com/tracking-iran-apt-screening-serpens/
logsource:
  product: windows
  category: file_event
detection:
  selection_ext:
    TargetFilename|endswith: '.exe.config'
  selection_terms:
    - '<appDomainManagerType'
    - 'AppDomainManager'
    - '<etwEnable enabled="false"'
    - '<bypassTrustedAppStrongNames enabled="true"'
    - '<publisherPolicy apply="no"'
    - '<probing privatePath="."'
    - '<probing privatePath="./"'
  condition: selection_ext and selection_terms
falsepositives:
  - Legitimate enterprise .NET applications with custom AppDomainManager usage. Baseline by signed publisher, installation directory, and change-management record.
level: high
tags:
  - attack.defense_evasion
  - attack.t1574
  - attack.t1218

```

6.3. Sigma detection: OpenSSH reverse tunnel pattern

```

title: Reverse SSH Tunnel With Remote Port Forwarding on Windows
id: 1ef4c9e4-unc1549-reverse-ssh-tunnel
status: experimental
description: Detects OpenSSH launched with remote port forwarding and options reported in UNC1549 intrusions.
author: Lost Boys Cyber
references:
  - https://cloud.google.com/blog/topics/threat-intelligence/analysis-of-unc1549-ttps-targeting-aerospace-defense
logsource:
  product: windows
  category: process_creation
detection:
  selection_img:
    Image|endswith:
      - '\ssh.exe'
      - '\OpenSSH\ssh.exe'
  selection_args:
    CommandLine|contains|all:
      - ' -R '
      - ' -N'
  suspicious_options:
    CommandLine|contains:
      - 'StrictHostKeyChecking=no'
      - 'UserKnownHostsFile=/dev/null'
      - 'ServerAliveInterval=60'
      - ' -p 443'
  condition: selection_img and selection_args and suspicious_options
falsepositives:
  - Administrator-approved reverse tunnels. Require ticket, owner, destination, and expected schedule.
level: high
tags:
  - attack.command_and_control
  - attack.t1090
  - attack.t1572

```

6.4. Sigma detection: UNC1549-style DLL sideloading near trusted vendor software

```

title: Suspicious DLL Sideloading Near Remote Access or Vendor Tools
id: 3c7d1d81-unc1549-sideloading-vendor-paths
status: experimental
description: Detects DLL creation/loading in directories and filename patterns associated with UNC1549
search-order hijacking tradecraft. Tune for local environment.
author: Lost Boys Cyber
references:
  - https://cloud.google.com/blog/topics/threat-intelligence/analysis-of-unc1549-ttps-targeting-aerospace-defense
  - https://cloud.google.com/blog/topics/threat-intelligence/suspected-iranian-unc1549-targets-israel-middle-east
logsource:
  product: windows
  category: image_load
detection:
  selection_dll:
    ImageLoaded|endswith:
      - '\secur32.dll'
      - '\VGAAuth.dll'
      - '\lastenzug.dll'
      - '\Lastenzug.dll'
      - '\CoreUIComponent.dll'
      - '\UpdateChecker.dll'
      - '\Updater.dll'
      - '\InitInstall.dll'
      - '\uevmonitor.dll'
      - '\unbcl.dll'
      - '\Connection.dll'
  suspicious_paths:
    ImageLoaded|contains:
      - '\Microsoft\OneDrive\configs\'
      - '\Microsoft\Internet Explorer\'
      - '\Microsoft\OneDrive\cache\logger\'
      - '\VMware\VMware Tools\VMware VGAAuth\'
      - '\AppData\Local\'
      - '\bin\update\'
  condition: selection_dll or suspicious_paths
falsepositives:
  - Legitimate DLL loads with the same names in expected signed vendor directories. Correlate with signer mismatch, new file creation, and rare parent processes.
level: medium
tags:
  - attack.defense_evasion
  - attack.t1574.001

```

6.5. KQL hunt: Azure Web Apps C2 from unusual processes

```

let suspicious_terms = dynamic([
  "azurewebsites.net", "cloudapp.azure.com", "onlyoffice.com", "filemail.com"
]);
let suspicious_processes = dynamic([
  "setup.exe", "update.exe", "FileCoAuth.exe", "SoftwareLicencing.exe", "ssh.exe",
  "VGAAuthCLI.exe", "BringThemeHome.exe", "BringThemeHomeNow.exe"
]);
DeviceNetworkEvents
| where Timestamp > ago(30d)
| where RemoteUrl has_any (suspicious_terms)
| extend Proc = tolower(InitiatingProcessFileName), Url = tolower(RemoteUrl)
| where Proc in~ (suspicious_processes)
  or InitiatingProcessFolderPath has_any ("appdata\\local", "downloads", "temp", "vmware", "onedrive\\cache", "bin\\update")
  or Url matches regex @"(hiring|career|air|ticket|survey|question|update|licen[cs]e|theme|health|finance|business|storage|vm|nx|connect).*\. (azurewebsites\.net|cloudapp\.azure\.com)"
| project Timestamp, DeviceName, InitiatingProcessAccountName, InitiatingProcessFileName,

```

```
InitiatingProcessCommandLine, InitiatingProcessFolderPath, RemoteUrl, RemoteIP, RemotePort
| order by Timestamp desc
```

6.6. KQL hunt: scheduled-task persistence created near suspicious archive execution

```
let suspicious_names = dynamic(["WindowsSecurityUpdate", "Synchronize OS", "OneDriveCoUpdate",
"OneDrive FileCoAuth", "Image Photo Viewer"]);
DeviceEvents
| where Timestamp > ago(30d)
| where ActionType in ("ScheduledTaskCreated", "ScheduledTaskUpdated")
| extend TaskName = tostring(parse_json(AdditionalFields).TaskName)
| where TaskName has_any (suspicious_names)
    or InitiatingProcessCommandLine has_any ("UpdateChecker.dll", "SoftwareLicensing.exe",
"FileCoAuth.exe", "bin\\update", "AppData\\Local")
| project Timestamp, DeviceName, InitiatingProcessAccountName, TaskName, InitiatingProcessFileName,
InitiatingProcessCommandLine, AdditionalFields
| order by Timestamp desc
```

6.7. KQL hunt: AppDomainManager config file creation and suspicious .NET sideloading

```
DeviceFileEvents
| where Timestamp > ago(30d)
| where FileName endswith ".exe.config"
| where FolderPath has_any ("Downloads", "Temp", "AppData\\Local", "bin\\update")
    or InitiatingProcessFileName in~ ("setup.exe", "powershell.exe", "cmd.exe", "wscript.exe",
"cscript.exe")
| join kind=leftouter (
    DeviceTvmSoftwareEvidenceBeta
    | summarize KnownSoftware=count() by DeviceId, FolderPath
) on DeviceId, FolderPath
| project Timestamp, DeviceName, InitiatingProcessAccountName, FileName, FolderPath,
InitiatingProcessFileName, InitiatingProcessCommandLine, SHA256
| order by Timestamp desc
```

If endpoint content indexing is available, search the created .config contents for `appDomainManagerType`, `etwEnable enabled="false"`, `bypassTrustedAppStrongNames`, `publisherPolicy apply="no"`, and `probing privatePath`.

6.8. KQL hunt: DCSync and suspicious domain-controller account manipulation

```
let lookback = 30d;
let dcsync_events = SecurityEvent
| where TimeGenerated > ago(lookback)
| where EventID == 4662
| where ObjectServer == "DS"
| where Properties has_any ("Replicating Directory Changes", "Replicating Directory Changes All", "DS-
Replication-Get-Changes")
| project TimeGenerated, Computer, Account, SubjectAccount, ObjectName, Properties, Activity;
let dc_account_changes = SecurityEvent
| where TimeGenerated > ago(lookback)
| where EventID in (4724, 4742, 4738)
| where TargetAccount endswith "$"
| project ChangeTime=TimeGenerated, Computer, Actor=SubjectAccount, TargetAccount, Activity;
dcsync_events
| join kind=leftouter dc_account_changes on $left.Account == $right.TargetAccount
| project TimeGenerated, Computer, Account, SubjectAccount, ObjectName, Properties, ChangeTime, Actor,
Activity1
| order by TimeGenerated desc
```

6.9. SPL hunt: reverse SSH and remote-access tooling

```
index=edr sourcetype=process
(Image="*\\ssh.exe" OR process_name="ssh.exe")
(CommandLine="* -R *" AND CommandLine="* -N*" AND (CommandLine="*StrictHostKeyChecking=no*" OR
CommandLine="*UserKnownHostsFile=/dev/null*" OR CommandLine="* -p 443*"))
| table _time host user parent_process process_name CommandLine dest_ip
| sort - _time
```

6.10. SPL hunt: UNC1549-style sideloaded DLL names

```
index=edr (sourcetype=image_load OR sourcetype=file_event)
(file_name IN
("secur32.dll","VGAAuth.dll","lastenzug.dll","Lastenzug.dll","CoreUIComponent.dll","UpdateChecker.dll","
Updater.dll","InitInstall.dll","uevmonitor.dll","unbcl.dll","Connection.dll"))
OR file_path="*\\Microsoft\\OneDrive\\cache\\logger\\"
OR file_path="*\\VMware\\VMware Tools\\VMware VGAAuth\\"
OR file_path="*\\bin\\update\\"
| stats values(file_path) as paths values(process_name) as processes values(parent_process) as parents
values(signature_status) as sig by host user file_name hash
| sort file_name
```

6.11. YARA: MiniUpdate / AppDomainManager loader configuration artifacts

```
rule LBC_UNC1549_AppDomainManager_Config_Indicators
{
  meta:
    author = "Lost Boys Cyber"
    description = "Detects text patterns in .NET configuration files associated with
AppDomainManager hijack and telemetry weakening observed in Screening Serpens / UNC1549-style 2026
tradecraft."
    reference = "https://unit42.paloaltonetworks.com/tracking-iran-apt-screening-serpens/"
    tlp = "CLEAR"
  strings:
    $cfg1 = "appDomainManagerType" ascii wide nocase
    $cfg2 = "appDomainManagerAssembly" ascii wide nocase
    $cfg3 = "<etwEnable enabled=\"false\"\" ascii wide nocase
    $cfg4 = "bypassTrustedAppStrongNames" ascii wide nocase
    $cfg5 = "publisherPolicy apply=\"no\"\" ascii wide nocase
    $cfg6 = "probing privatePath" ascii wide nocase
    $cfg7 = "requiredRuntime safemode=\"true\"\" ascii wide nocase
  condition:
    uint16(0) != 0x5a4d and filesize < 2MB and 3 of ($cfg*)
}
```

6.12. YARA: DCSYNCR.SLICK-inspired utility markers

```
import "pe"

rule LBC_UNC1549_DCSYNCR_SLICK_Utility_Markers
{
  meta:
    author = "Lost Boys Cyber"
    description = "Detects credential-dumping utility strings and LOG.txt output patterns
consistent with public DCSYNCR.SLICK reporting."
    reference = "https://cloud.google.com/blog/topics/threat-intelligence/analysis-of-unc1549-ttps-
targeting-aerospace-defense"
    tlp = "CLEAR"
  strings:
    $log1 = "\\LOG.txt" ascii wide
    $fmt1 = "%ws_%d:%d:" ascii wide fullword
    $fmt2 = "%ws:%d:" ascii wide fullword
    $sep1 = "::::" ascii wide fullword
    $path1 = "C:\\users\\public\\LOG.txt" ascii wide nocase
    $path2 = "VMware\\VMware Tools\\VMware VGAAuth\\LOG.txt" ascii wide nocase
  condition:
    pe.is_pe and filesize < 10MB and $log1 and 2 of ($fmt*, $sep1, $path*)
}
```

6.13. Analyst hunting playbook

Hunt	Steps	Escalate if
Recruitment lure execution	Search email/proxy for job, hiring, career, meeting, ONLYOFFICE, filemail, and Azure Web App links; pivot to endpoint	Archive contains hidden DLLs, unsigned DLLs, nested ZIPs, or execution from

	archive extraction and setup.exe execution.	Downloads/Temp.
Sideloaded	Find signed binaries loading new DLLs from writable directories; compare DLL signer and creation time; inspect parent process and command line.	DLL created shortly before load, unsigned/mismatched signer, user-writable path, or known UNC1549 filename.
AppDomainManager hijack	Inventory .exe.config files containing AppDomainManager or ETW disabling directives; validate business owner.	Config resides outside a managed install path, appeared after user download, or references local attacker-controlled DLL.
Reverse tunnels	Hunt ssh.exe with <code>`-R`, `-N`, `StrictHostKeyChecking=no`, `UserKnownHostsFile=/dev/null`, TCP/443</code> ; enrich destination IP/domain.	Destination is rare, non-corporate, cloud/VPS, or launched by user/service account without approved admin workflow.
DCSync / AD abuse	Check 4662 replication events, DC computer-account password resets, new computer accounts, RBCD changes, certificate enrollments, Kerberoast spikes.	Replication performed by non-DC, new computer account, manipulated DC account, or unexpected admin workstation.
Third-party VDI breakout	Review child processes spawned inside virtual sessions, file transfers, mapped drives, clipboard use, and outbound scanning.	Vendor account launches shell, PowerShell, browser credential access, network scanner, or remote-control tooling.

6.14. Detection engineering notes

- Build allowlists cautiously. UNC1549 abuses legitimate tools and legitimate-looking paths; a signed parent process is not enough to suppress an alert.
- Treat Azure Web App domains as context-rich rather than intrinsically malicious. Score by process, user, domain age/rarity, path, business need, and similarity to job / airline / engineering / service-ticket themes.
- Add a content-inspection control for .NET .config files in user-writable directories. This is a low-volume artifact class with high signal when it disables ETW or declares custom AppDomainManager entries.
- Combine identity and endpoint telemetry. For example, DCSync after a vendor VDI login or after ssh.exe reverse-tunnel creation is a higher-confidence incident than either signal alone.
- After remediation, maintain a 60-90 day watch for dormant beacons, rare scheduled tasks, and cloud C2 callbacks. Mandiant reported UNC1549 backdoors that remained quiet until post-eradication.

7. Recommendations

Priority	Recommendation	Owner	Implementation guidance	Expected outcome
1	Harden third-party access and VDI boundaries.	Identity / Infrastructure	Enforce phishing-resistant MFA, conditional access, device compliance, session recording for high-risk vendors, no direct admin paths from partner VDI, restricted clipboard/file transfer, and segmented partner networks.	Reduces UNC1549's highest-leverage access path.
2	Deploy sideloading and	SOC / Detection	Implement the	Detects both historical

	AppDomainManager detection.	Engineering	Sigma/KQL analytics above; baseline known enterprise .NET AppDomainManager use; alert on new .exe.config files in user-writable paths.	and 2026 execution tradecraft.
3	Monitor and restrict outbound tunnels.	Network / EDR	Alert on ssh.exe reverse tunnels, ngrok, ZeroTier, unusual SOCKS/WebSocket tunnels, and outbound TCP/443 from non-browser admin tools. Require approved tunnel inventory.	Reduces covert C2 and post-remediation persistence.
4	Improve identity-attack detection for DCSync, RBCD, AD CS, and DC-account manipulation.	Identity / Active Directory	Audit replication rights, alert on 4662 replication, Event ID 4742/4724 changes to computer accounts, AD CS ESC template exposure, Kerberoasting spikes, and new machine-account abuse.	Detects credential escalation and domain compromise attempts.
5	Add recruitment-lure controls for high-risk personnel.	Security Awareness / Email	Flag external recruiter messages with archives, nested ZIPs, ONLYOFFICE/file-sharing links, Azure-hosted landing pages, or job portals asking for credentials. Provide secure detonation workflow for job documents.	Reduces initial execution and credential harvesting risk.
6	Tighten cloud and proxy controls around rare Azure Web Apps.	Network / Cloud Security	Use DNS/proxy risk scoring for rare azurewebsites.net and cloudapp.azure.com domains, especially with job, airline, survey, support, ticket, update, health, finance, VM, or licensing terms.	Surfaces cloud-C2 patterns without blocking all Azure.
7	Prepare an UNC1549 incident-response eradication checklist.	IR / SOC	Include full credential reset, partner tokens, Azure AD Connect credentials, browser session revocation, scheduled task review, VDI logs, reverse-tunnel sweeps, and 90-day	Reduces chance of post-remediation re-entry.

			dormant beacon watch.	
8	Segment sensitive engineering and defense data repositories.	Engineering / IT	Restrict SharePoint, Teams, Git, PLM, ticketing, and build-system access by role and device posture; alert on abnormal bulk access and external partner reads.	Limits espionage value after account compromise.

Executive actions:

- Require a board-level owner for supplier remote-access risk. UNC1549's trusted-relationship tradecraft turns partner governance into a core security control.
- Fund identity telemetry before buying more malware signatures. The actor's decisive moves are often account-based: vendor credentials, password resets, DCSync, RBCD, AD CS, and browser credential theft.
- Treat recruitment and meeting workflows as attack surfaces. High-value technical employees should have a safe path to submit job/recruiter documents for detonation without stigma.
- Conduct a tabletop exercise around a supplier account entering Citrix/AVD, breaking out, staging a reverse SSH tunnel, and attempting DCSync. This scenario closely matches public reporting and will reveal cross-team gaps.

Immediate 14-day action plan:

Day range	Action
0-2	Load public UNC1549 / Screening Serpens IOCs into SIEM as enrichment-only watchlists; do not rely on them as sole blocking logic.
0-5	Deploy reverse SSH and AppDomainManager detection in audit mode; review false positives with IT operations.
3-7	Hunt for suspicious scheduled tasks, sideloaded DLLs, and rare Azure Web Apps traffic from user-writable paths over the prior 90 days.
5-10	Review vendor / partner accounts with VDI, Citrix, VMware, AVD, VPN, and admin-adjacent access; enforce conditional access and session logging.
7-14	Audit AD replication rights, AD CS templates, RBCD assignments, and Azure AD Connect credentials; remediate excessive privilege.

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